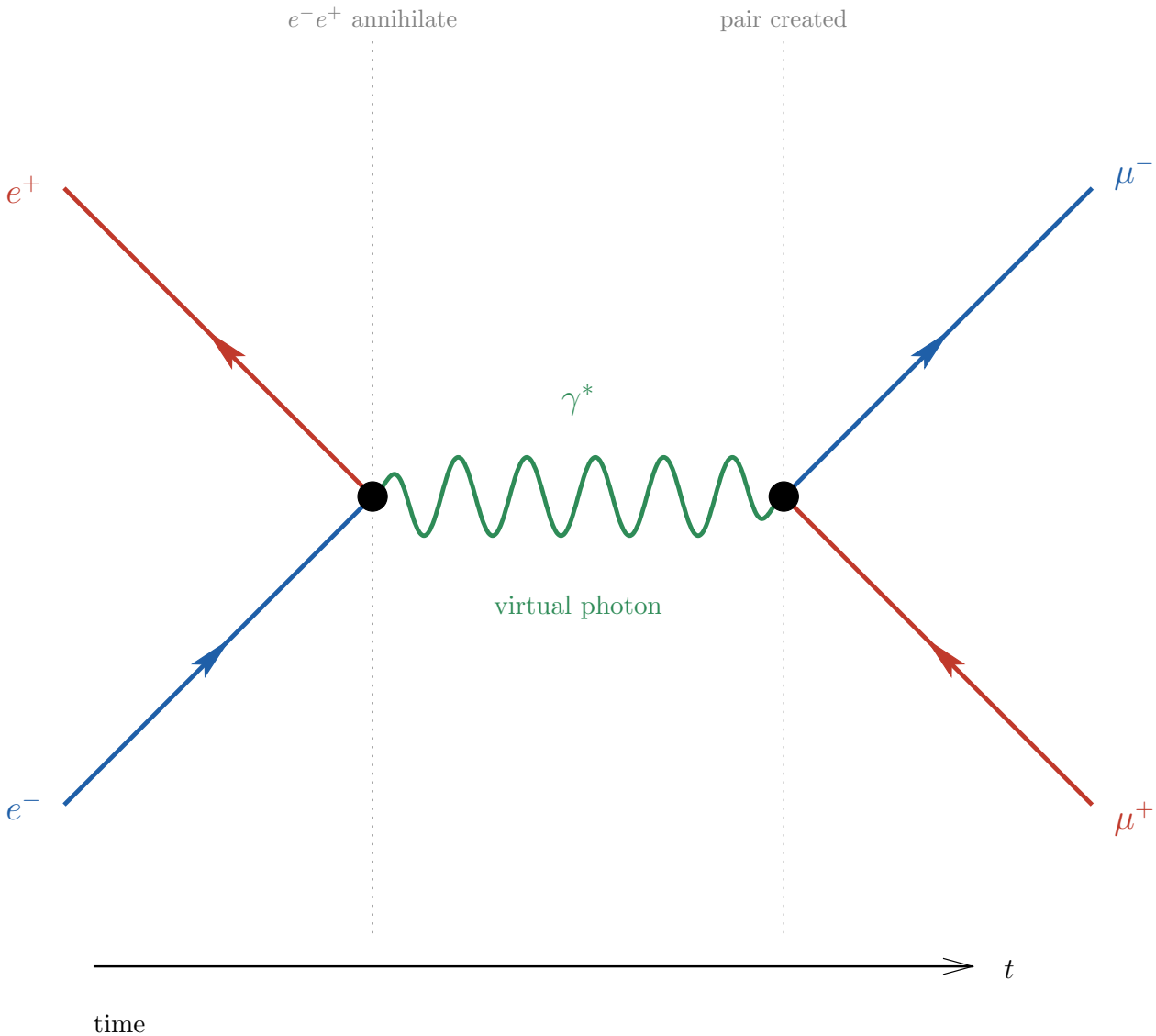
Electron-positron annihilation into muons

Lowest-order QED process, time flows to the right



Process

$e^- e^+ \to \gamma^* \to \mu^- \mu^+$

Arrow: flow of lepton number; the antiparticle's arrow points backwards in time.

Cross section (QED):

$\sigma = 4\pi\alpha^2/3s$